

Assignment 3

Case A

1) Model Formulation

Sets and Parameters

- Let H denote Houston (the hub), $S = C, M, P$ denote the three spokes: Chicago (C), Miami (M), Phoenix (P).
- Let $I = H \cup S$ be the set of all cities.
- Indices $i, j \in I$ with $i \neq j$. A route $i \rightarrow j$ is a one-way ticket from origin i to destination j . (We exclude $i = j$.)
- p_{ij} = price charged for a one-way coach ticket on route $i \rightarrow j$ (from Table 1).
- d_{ij} = (known) demand for route $i \rightarrow j$ during one bank (from Table 2).
- C = aircraft capacity in coach seats on each flight (given as 240).
- Marginal cost per passenger ≈ 0 , so objective maximizes revenue.

Decision variables

For every ordered pair $i \neq j$:

x_{ij} ; =, number of tickets sold for route $i \rightarrow j$.

(Require integer $x_{ij} \geq 0$; relaxation to real numbers gives an LP.)

Flight-leg capacity mapping (which routes occupy which flights)

Each bank consists of six flights (three inbound spokes $\rightarrow H$, and three outbound $H \rightarrow$ spokes). A passenger ticket $i \rightarrow j$ uses:

- If $i \in S$ and $j = H$: the inbound flight $i \rightarrow H$.
- If $i = H$ and $j \in S$: the outbound flight $H \rightarrow j$.

- If $i \in S$ and $j \in S, j \neq i$: a two-leg itinerary: first leg $i \rightarrow H$, second leg $H \rightarrow j$. Therefore such a passenger consumes one seat on both the inbound flight from i and the outbound flight to j .

Objective

Maximize total revenue for the bank:

$$\max Z = \sum_{i \in I} \sum_{j \in I, j \neq i} p_{ij} x_{ij}.$$

Constraints

1. **Demand bounds (cannot sell more than demand):**

$$0 \leq x_{ij} \leq d_{ij} \quad \forall, i \neq j.$$

2. **Inbound leg capacity** for each spoke $s \in S$ (flight $s \rightarrow H$ capacity C): all passengers originating at s use this leg:

$$x_{sH} + \sum_{t \in S, t \neq s} x_{st} \leq C$$

$$\forall, s \in C, M, P.$$

(Left side = tickets with origin s : direct-to-hub plus two-leg departures.)

3. **Outbound leg capacity** for each spoke $s \in S$ (flight capacity C): all passengers whose second leg is $H \rightarrow s$ use this leg:

$$x_{Hs} + \sum_{t \in S, t \neq s} x_{ts} \leq C$$

$$\forall, s \in C, M, P.$$

(Left side = tickets whose destination is s : direct-from-hub plus connecting arrivals destined for s .)

4. **Nonnegativity:**

$$x_{ij} \in \mathbb{Z}_{\geq 0} \quad \forall, i \neq j.$$

(If integer requirement is relaxed, the problem is a linear program.)

2) & 3)

Optimal bookings per route

Decision Variables: Tickets sold for each route					
		Destination			
		Houston	Chicago	Miami	Phoenix
Origin	Houston	0.00	123.00	80.00	110.00
	Chicago	130.00	0.00	94.00	16.00
	Miami	72.00	100.00	0.00	68.00
	Phoenix	115.00	17.00	66.00	0.00

Optimal Revenue: **100568**

Case B

Constraints

Cell	Name	Final Value	Shadow Price	Constraint R.H. Side	Allowable Increase	Allowable Decrease
\$D\$35	C → H Capacity LHS	240	88	240	46	16
\$D\$36	M → H Capacity LHS	240	15	240	5	42
\$D\$37	P → H Capacity LHS	198	0	240	1E+30	42
\$D\$38	H → C Capacity LHS	240	90	240	42	17
\$D\$39	H → M Capacity LHS	240	10	240	4	46
\$D\$40	H → P Capacity LHS	194	0	240	1E+30	46

Sensitivity Report

1)

They should accept the offer.

The sensitivity report shows that the shadow price of M→H is \$15, which is lower than the offer they provides, and it is within the allowable increase. So they should take the offer.

And we resolve the case to test it(as showed in sheet 'Case B-1 test'): Lower the constraint of $M \rightarrow H$ to 235 and add another 5 seats on Miami to Houston, the optimal revenue is 105337, greater than the Case A value.

Decision Variables: Tickets sold for each route					
		Destination			
		Houston	Chicago	Miami	Phoenix
Origin	Houston	0.00	123.00	80.00	110.00
	Chicago	130.00	0.00	94.00	16.00
	Miami	77.00	95.00	0.00	68.00
	Phoenix	115.00	22.00	66.00	0.00
Objective Function					
Total Revenue		100853			
Constraints		LHS		RHS	
C $\rightarrow$ H Capacity		240.00		240	
M $\rightarrow$ H Capacity		240.00		235	
P $\rightarrow$ H Capacity		203.00		240	
H $\rightarrow$ C Capacity		240.00		240	
H $\rightarrow$ M Capacity		240.00		240	
H $\rightarrow$ P Capacity		194.00		240	
Demand		D25:G28		D15:G18	

2)

They should accept the offer.

The (shadow price $M \rightarrow H$ + shadow price $H \rightarrow C$) is less than \$285, so they should accept the offer.

We use the same testing method as 1) did, the optimal revenue goes higer.

Case C

1)

They should purchase 240 - seats aircraft to minimize cost.

Decision Variables: Tickets sold for each route					
		Destination			
		Houston	Chicago	Miami	Phoenix
Origin	Houston	0.00	123.00	80.00	110.00
	Chicago	130.00	0.00	94.00	16.00
	Miami	72.00	100.00	0.00	68.00
	Phoenix	115.00	17.00	66.00	0.00
Seat Capacity		240			
Objective Function					
Total Revenue		-29032			
Constraints		LHS		RHS	
C → H Capacity		240.00		240	
M → H Capacity		240.00		240	
P → H Capacity		198.00		240	
H → C Capacity		240.00		240	
H → M Capacity		240.00		240	
H → P Capacity		194.00		240	
Demand		D25:G28		D15:G18	

2)

1)

Yes, they should be allocated with same routes. If demand patterns are stable across banks, it's efficient to assign aircraft consistently to the same pair of routes (e.g., one aircraft always does Chicago–Houston + Houston–Chicago)

2)

Decision Variables: Tickets sold for each route							
		Destination					
		Houston	Chicago	Miami	Phoenix		
Origin	Houston	0.00	123.00	80.00	110.00		
	Chicago	130.00	0.00	98.00	88.00		
	Miami	72.00	105.00	0.00	68.00		
	Phoenix	115.00	90.00	66.00	0.00		
		C	M	P			
Leg Capacity		318	245	271	318	245	271
Total Cost		12318	12245	12271			
Objective Function							
Total Revenue less c		40723.00					
Constraints		LHS		RHS			
C → H Capacity		316.00		318			
M → H Capacity		245.00		245			
P → H Capacity		271.00		271			
H → C Capacity		318.00		318			
H → M Capacity		244.00		245			
H → P Capacity		266.00		271			
Demand		D25:G28		D15:G18			

According to the solver solution, they should purchase new aircrafts with **318 seats, 245 seats and 271 seats** accordingly.

3)

They shouldn't take that discount.

BlueSky operates 3 banks per weekday, 5 weekdays per week, 52 weeks per year.
That's:

$$\text{Banks per year} = 3 \cdot 5 \cdot 52 = 780$$

The \$5,000,000 one-time discount spread across this horizon gives a per-bank value:

$$\text{Discount per bank} = \frac{5,000,000}{780} \approx 6,410.26$$

Adding the discount, the profit of purchasing 3 identical aircraft is $-29032 + 6140.26 = -22621.74$, which is still less than the profit of purchasing different aircraft.